

Task 1 – Weather API Integration

Connect to a Weather API to fetch current weather details for a given city.

Instructions:

- Prompt AI to generate Python code using the requests library.
- Handle errors: invalid city name, API key errors, and network failures.
- Display temperature, humidity, and weather description in a readable format.

Expected Output:

- A working Python script that fetches and displays weather data with proper error handling.

Code :

```
week 18 > demo.py > ...
1  """
2  Weather Fetcher
3  use open meteo API and geocoding API
4  """
5  import requests
6  def get_coordinates(location):
7      """Get the coordinates of a location using the geocoding API"""
8      url = f"https://geocoding-api.open-meteo.com/v1/search?name={location}"
9      response = requests.get(url)
10     data = response.json()
11     if "results" in data and len(data["results"]) > 0:
12         return data["results"][0]["latitude"], data["results"][0]["longitude"]
13     else:
14         raise ValueError("Location not found")
15 def get_weather(latitude, longitude):
16     """Get the current weather for the given coordinates using the Open Meteo API"""
17     url = f"https://api.open-meteo.com/v1/forecast?latitude={latitude}&longitude={longitude}&current_weather=true"
18     response = requests.get(url)
19     data = response.json()
20     if "current_weather" in data:
21         return data["current_weather"]
22     else:
23         raise ValueError("Weather data not found")
24 def main():
25     location = input("Enter a location: ")
26     try:
27         latitude, longitude = get_coordinates(location)
28         weather = get_weather(latitude, longitude)
29         print(f"Current weather in {location}:")
30         print(f"Temperature: {weather['temperature']}°C")
31         print(f"Wind Speed: {weather['windspeed']} km/h")
32         print(f"Weather Code: {weather['weathercode']}")
```

Output :

```
PS C:\Users\User\OneDrive\Desktop> python IAC/week 18/demo.py
Enter a location: warangal
Current weather in warangal:
Temperature: 33.3°C
Wind Speed: 1.3 km/h
Weather Code: 2
```

Task 2 – Currency Exchange Rates

Integrate a Currency Exchange API to convert INR to USD, EUR, and GBP.

Instructions:

- Use AI to generate API request code.
- Validate user input for amount and currency codes.
- Implement error handling for invalid responses and API downtime.
- Show conversion results clearly in tabular format.

Expected Output:

- A script that converts INR to multiple currencies with graceful error handling.

Code:

```
week 18 > currency.py > ...
1  """
2  design a currency exchanger that can convert inr to different currencies and vice versa.
3  use currency exchange api
4  """
5  import requests
6  def get_exchange_rate(from_currency, to_currency):
7      """Get the exchange rate from one currency to another using the Currency Exchange API"""
8      url = f"https://api.exchangerate-api.com/v4/latest/{from_currency}"
9      response = requests.get(url)
10     data = response.json()
11     if "rates" in data and to_currency in data["rates"]:
12         return data["rates"][to_currency]
13     else:
14         raise ValueError("Currency not found")
15  def convert_currency(amount, from_currency, to_currency):
16      """Convert an amount from one currency to another using the exchange rate"""
17      exchange_rate = get_exchange_rate(from_currency, to_currency)
18      return amount * exchange_rate
```

Output:

```

PS C:\Users\User\OneDrive\Desktop\AIAC> & C:/Users/User/OneDrive/Desktop/AIAC/week 18/currency.py
Enter the amount to convert: 100
Enter the currency to convert from (e.g., INR): INR
Enter the currency to convert to (e.g., USD): USD
100.0 INR is equal to 1.06 USD

```

Task 3 – GitHub Repository Info Fetcher

Connect to the GitHub API to fetch details of a repository (e.g., stars, forks, issues).

Instructions:

- Prompt AI to write a Python function to send GET requests to GitHub API.
- Handle rate limit errors, 404 (repo not found), and invalid input.
- Display repository name, description, stars, forks, and open issues.

Expected Output:

- A Python program that retrieves and displays GitHub repository information with robust error handling.

```

week 18 > github.py > main
4 write a Python function to send GET requests to GitHub API
5 """
6 import requests
7 def get_repo_info(owner, repo):
8     """Get the details of a GitHub repository using the GitHub API"""
9     url = f"https://api.github.com/repos/{owner}/{repo}"
10    response = requests.get(url)
11    if response.status_code == 200:
12        data = response.json()
13        return {
14            "name": data["name"],
15            "owner": data["owner"]["login"],
16            "stars": data["stargazers_count"],
17            "forks": data["forks_count"],
18            "open_issues": data["open_issues_count"]
19        }
20    else:
21        raise ValueError("Repository not found")

```

Output:

```
PS C:\Users\User\OneDrive\Desktop\AIAC> & C:/U
IAC/week 18/github.py"
Enter the repository owner: 2303a51019
Enter the repository name: AIAC_LAB
Repository: AIAC_LAB
Owner: 2303a51019
Stars: 0
Forks: 0
Open Issues: 0
PS C:\Users\User\OneDrive\Desktop\AIAC> █
```

Task 4 – Real-Time Application: News Headlines Aggregator

Scenario: Build a News Aggregator using a free News API.

Requirements:

- Fetch top 5 headlines for a given category (sports, technology, health).
- Handle errors like invalid category, missing API key, and request failures.
- Display headlines in a user-friendly numbered list format.
- Include a retry mechanism if the first request fails.

Expected Output:

- A working Python script that retrieves and displays news headlines with proper error handling

Code:

```

week 18 > news.py > ...
1  """
2  News Headlines Fetcher
3  use newsapi.org
4  """
5  import requests
6  def get_news_headlines(category, api_key):
7      """Fetch top 5 news headlines for a given category using the NewsAPI"""
8      url = f"https://newsapi.org/v2/top-headlines?category={category}&apiKey={api_key}"
9      response = requests.get(url)
10     if response.status_code == 200:
11         data = response.json()
12         if "articles" in data:
13             return [article["title"] for article in data["articles"][:5]]
14         else:
15             raise ValueError("No articles found")
16     elif response.status_code == 401:
17         raise ValueError("Invalid API key")
18     elif response.status_code == 400:
19         raise ValueError("Invalid category")

```

Output:

```

PS C:\Users\User\OneDrive\Desktop\AIAC> & C:/Users/User/AppData/Local/Python/pythoncore-3.14-64/python.exe "c:/Users/User/OneDrive/Desktop/AIAC/week 18/news.py"
Enter your NewsAPI key: 71b8ebd4f2f2456191db37f7faf6413a
Enter the news category (sports, technology, health): sports
Top 5 headlines in sports:
1. 2026 MLB predictions: Playoffs, World Series, MVPs and more - ESPN
2. Miami Open: Coco Gauff reaches semi-finals for first time - BBC
3. Cavs coach Kenny Atkinson sends pointed message to team about playoff fate following win over Orlando - Cleveland.com
4. Team USA's choice for Olympic flag football is obvious - theScore.com
5. Bryan Hodgson's wife stole the March Madness spotlight with shocking comment from new Providence coach - New York Post

```